



EAST WEST UNIVERSITY

Department of Computer Science & Engineering

Semester: Spring 2025

Course Code: CSE405

Course Title: Computer Networks

Section: 04

Project Report

Topic : Data Communication in a Network

Submitted to

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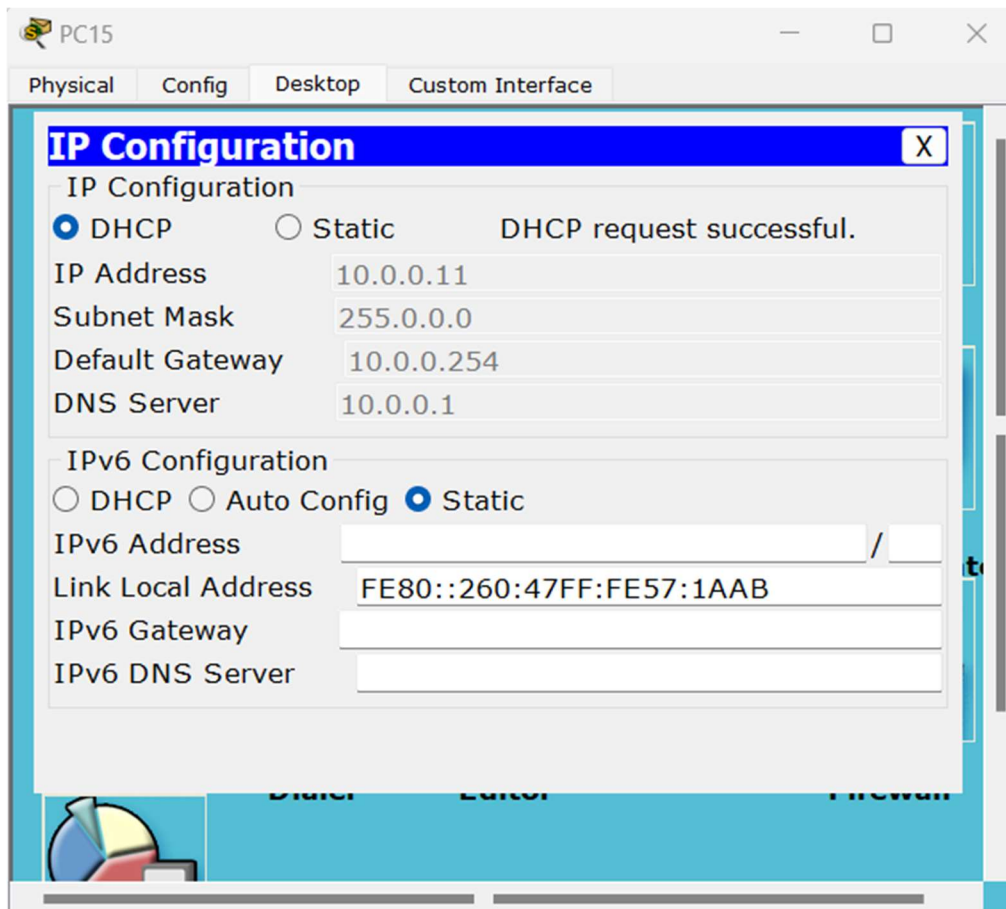
This project addresses the expansion of a corporate network infrastructure across five departmental buildings: Headquarters, Sales Office, Tech Center, Human Resources, and Finance Division. Using Cisco Packet Tracer, the network was developed with Class A IP addressing, OSPF routing for dynamic connectivity, and DHCP for automated address allocation. A centralized DNS and HTTP server was placed at the Headquarters to simulate critical business operations — including employee onboarding, asset management, and invoice processing — through locally hosted web interfaces. The goal is to create a scalable, functional, and service-oriented corporate network environment.

The diagram illustrates a multi-branch enterprise network. The central Tech Center (IP: 30.0.0.0, Gateway: 30.0.0.254) acts as the core, connecting to several branches via a central router (Router2, IP: 45.0.0.0, Gateway: 45.0.0.254). The branches include:

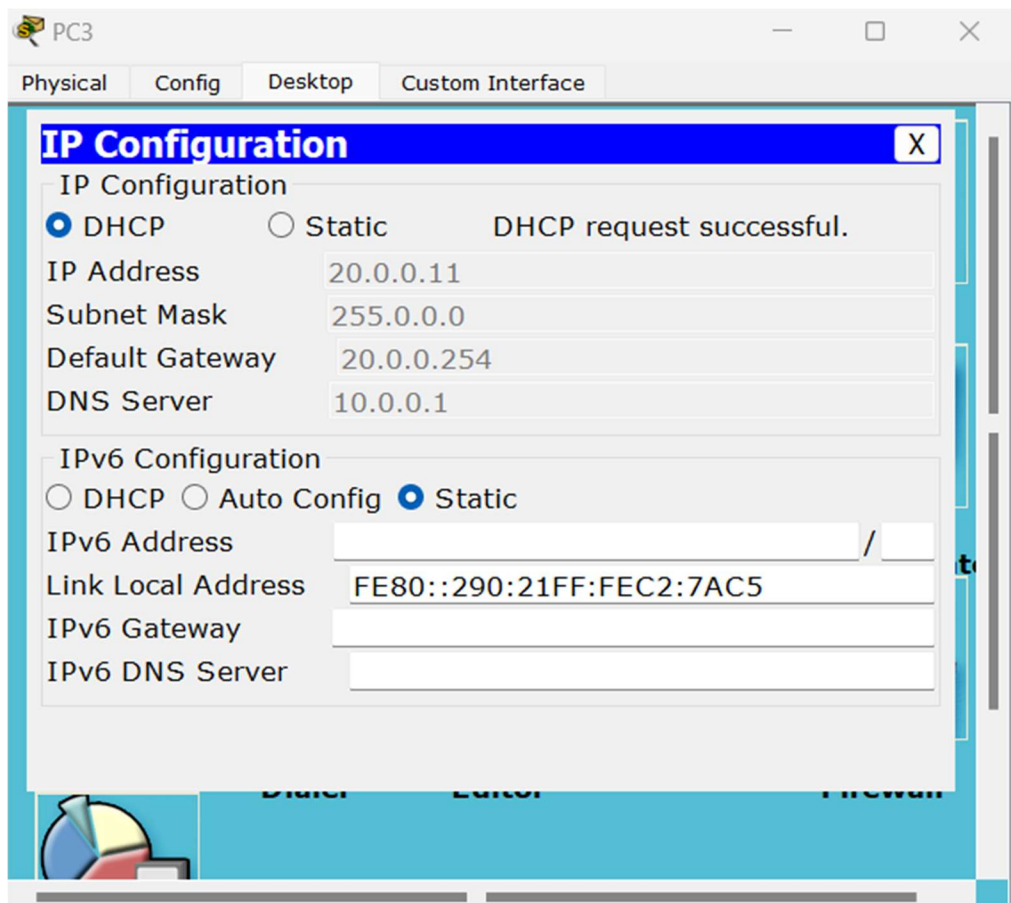
- Headquarters** (IP: 10.0.0.0, Gateway: 10.0.0.254): Connected to Router1 (IP: 15.0.0.0, Gateway: 15.0.0.254). Includes PC-PT PC15, PC-PT PC0, Server-PT Server0, Server-PT Server1, Server-PT Server2, Server-PT Server3, Server-PT Server4, Server-PT Server25, Web Server, DNS Server, and DHCP Server 1 through 5.
- Sales Office** (IP: 20.0.0.0, Gateway: 20.0.0.254): Connected to Router3 (IP: 35.0.0.0, Gateway: 35.0.0.254). Includes PC-PT PC3, PC-PT PC4, PC-PT PC5, Server-PT Server6, Server-PT Server7, Server-PT Server8, Server-PT Server9, Server-PT Server27, and DHCP Server 1 through 5.
- Human Resources** (IP: 40.0.0.0, Gateway: 40.0.0.254): Connected to Router4 (IP: 65.0.0.0, Gateway: 65.0.0.254). Includes PC-PT PC9, PC-PT PC10, PC-PT PC11, Server-PT Server15, Server-PT Server16, Server-PT Server17, Server-PT Server18, Server-PT Server28, and DHCP Server 1 through 5.
- Finance Division** (IP: 50.0.0.0, Gateway: 50.0.0.254): Connected to Router5 (IP: 85.0.0.0, Gateway: 85.0.0.254). Includes PC-PT PC12, PC-PT PC13, PC-PT PC14, Server-PT Server20, Server-PT Server21, Server-PT Server22, Server-PT Server23, Server-PT Server30, and DHCP Server 1 through 5.

The network is interconnected via a central Tech Center and various routers and switches. Red lines indicate specific network paths or connections.

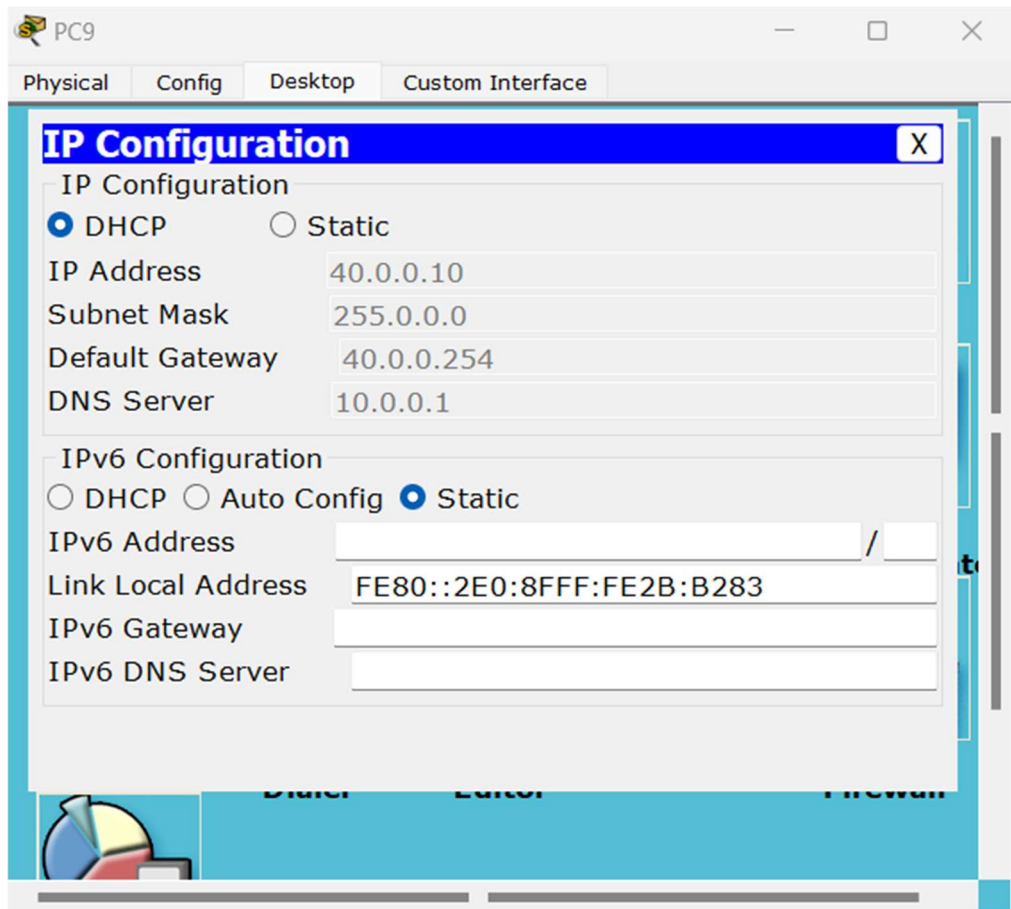
Experimental Results:



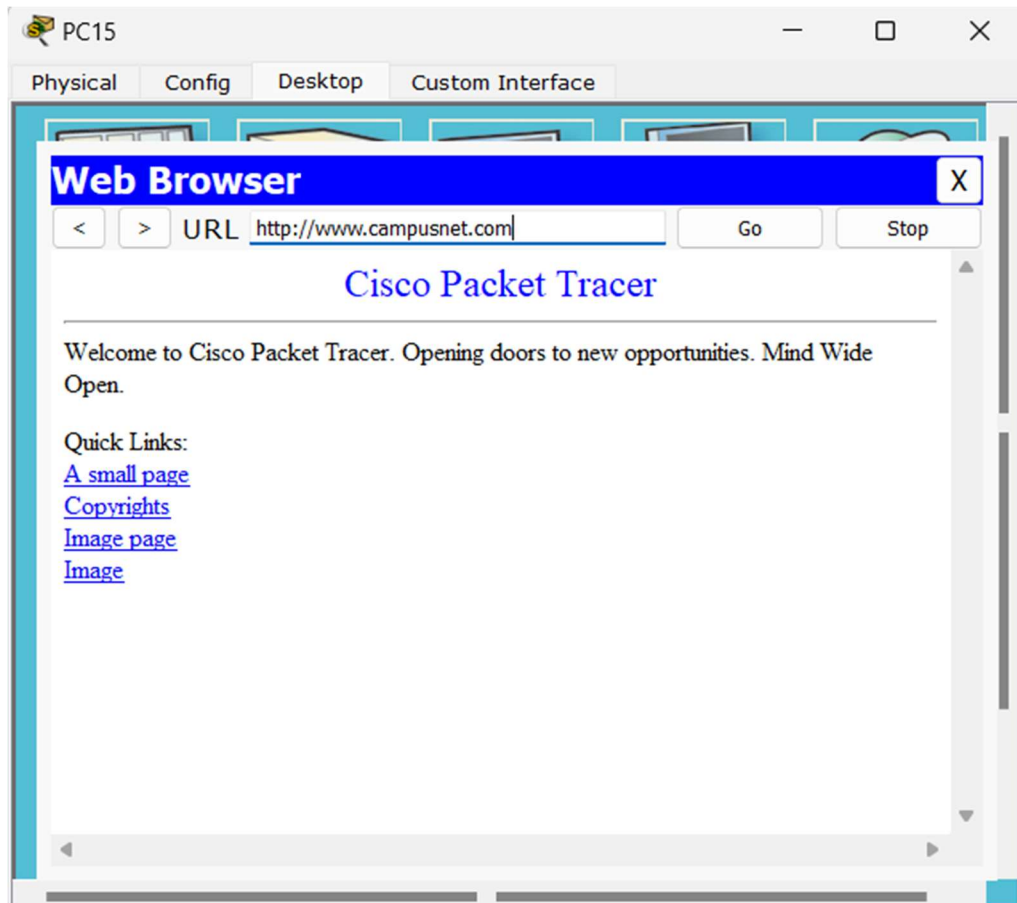
DHCP request successful in Headquarters



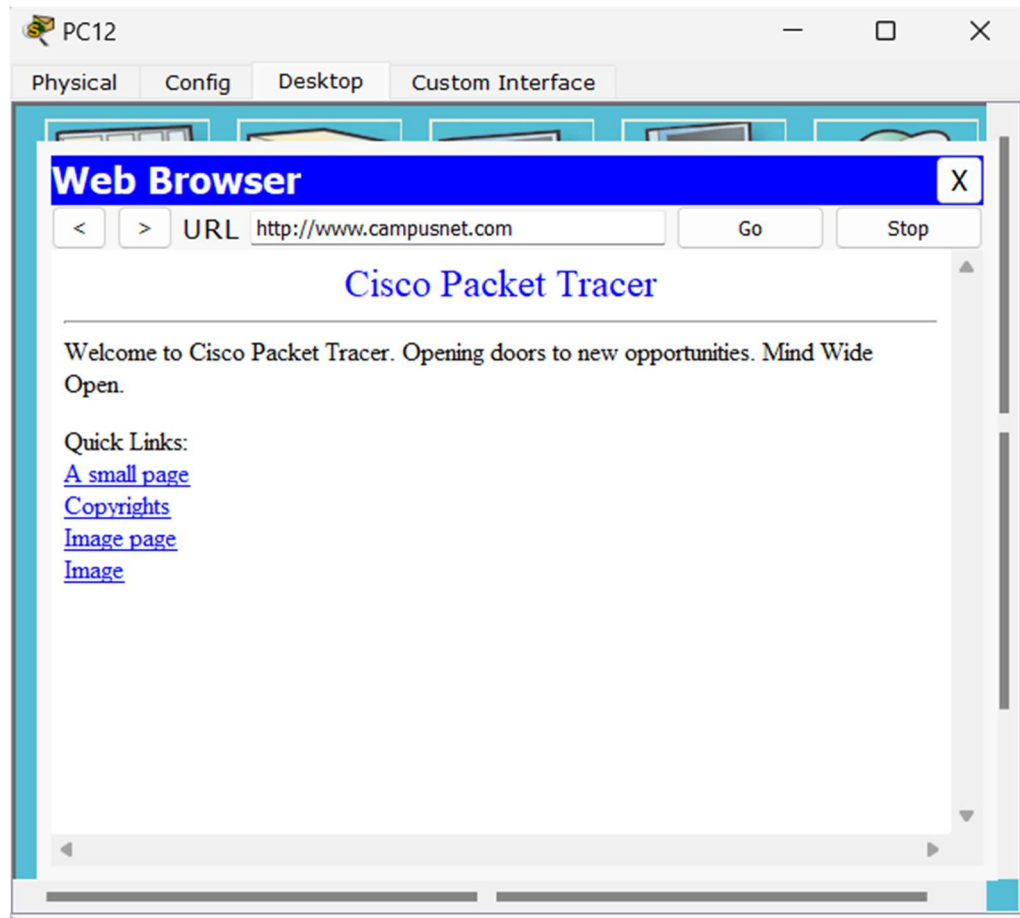
DHCP request successful in Sales Office



DHCP request successful in Human Resource



Web Browser giving the url from headquarters pc



Web Browser giving the url from other building pc also (Finance Division)

Server25

Physical Config Services Desktop Custom Interface

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.254

DNS Server: 10.0.0.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::2D0:BAFF:FE64:8794

IPv6 Gateway:

IPv6 DNS Server:

DNS Server

Server26

Physical Config Services Desktop Custom Interface

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.254

DNS Server: 10.0.0.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

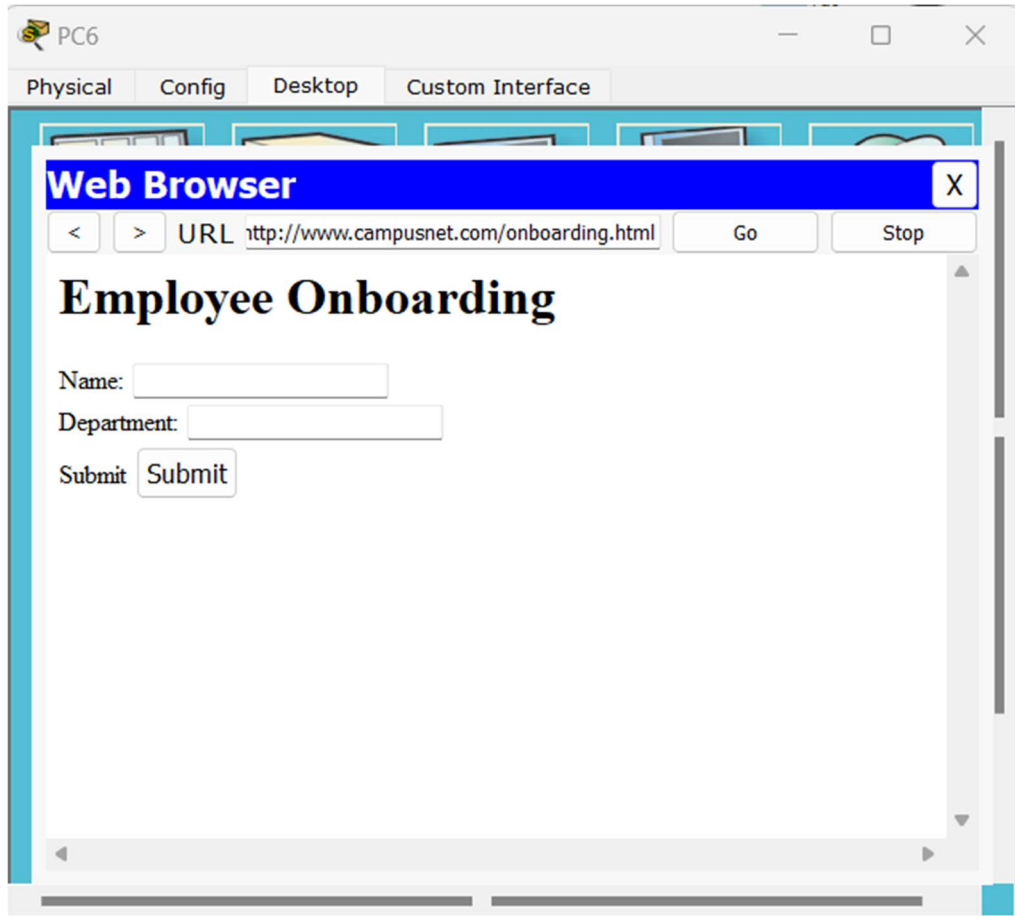
IPv6 Address: /

Link Local Address: FE80::260:70FF:FEB8:24E

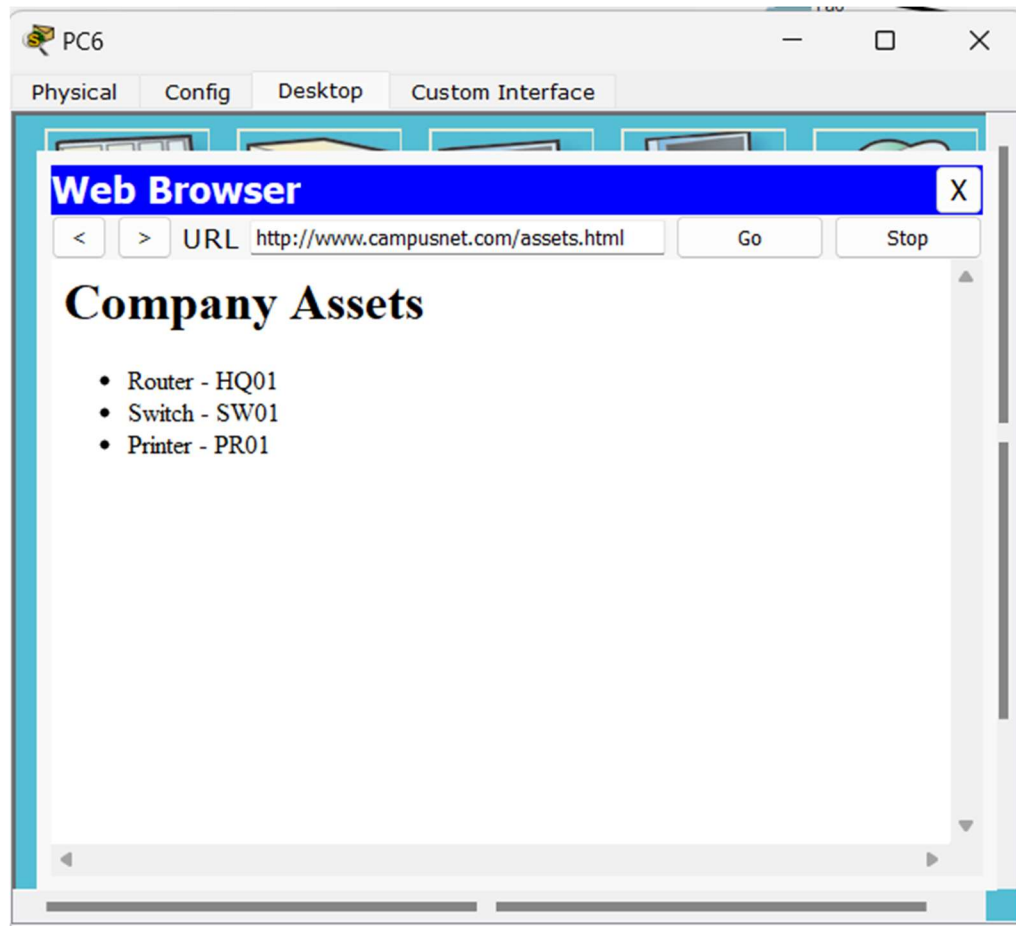
IPv6 Gateway:

IPv6 DNS Server:

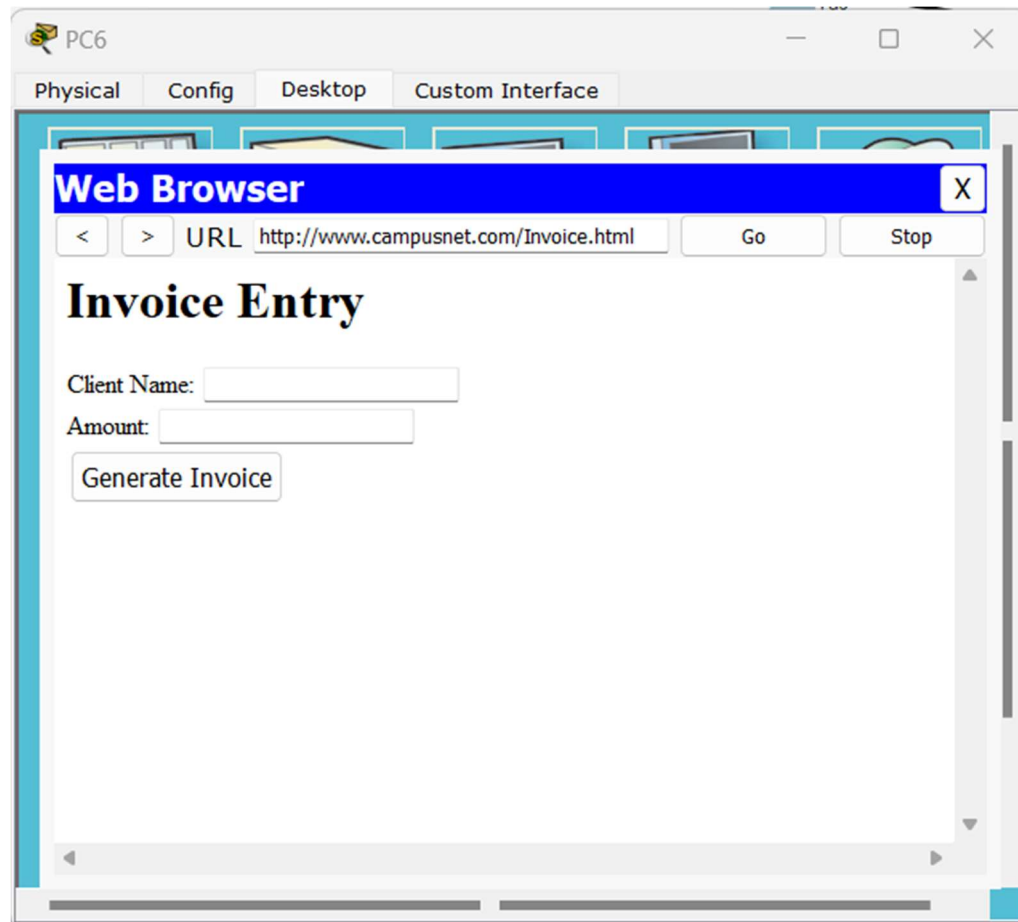
Web Server



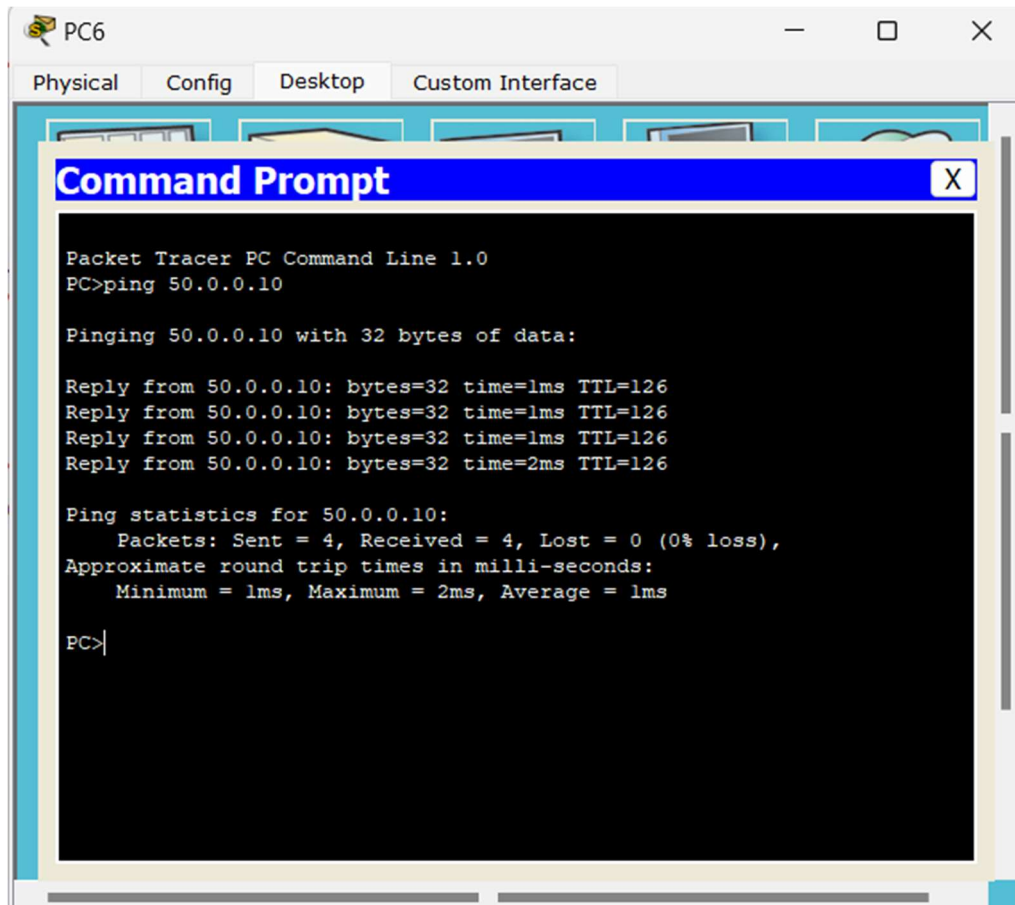
Employee onboarding



Assets.html



Invoice.html

The image shows a screenshot of a Packet Tracer PC Command Prompt window. The window title is "PC6" and it has tabs for "Physical", "Config", "Desktop", and "Custom Interface". The Command Prompt window has a blue title bar with the text "Command Prompt" and a close button. The text inside the Command Prompt window is as follows:

```
Packet Tracer PC Command Line 1.0
PC>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=2ms TTL=126

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

PC>|
```

Ping test from Tech Center Pc to Finance Division Pc successful

Conclusion:

The network simulation successfully fulfilled the requirements of a modern corporate infrastructure by ensuring dynamic routing, centralized services, and automation across multiple buildings. The implementation of OSPF routing and DHCP services enabled seamless connectivity and efficient IP management. Additionally, simulating business processes through HTML pages hosted on the internal web server provided a practical demonstration of how IT services can support real-life operations in a corporate setting. This project reflects both technical competence and an understanding of how networking supports enterprise functionality.